WEI CHEN

Ph.D. Candidate | Electrical and Computer Engineering | Purdue University
Email: chen2732@purdue.edu | Website: weichennone.github.io/myhomepage

RESEARCH INTERESTS

Understanding how knowledge is represented, organized, and composed within neural networks to build AI systems that can efficiently learn, remember, retrieve, and combine skills over time.
Research Area: Deep learning, computer vision, representation learning, and lifelong learning.

EDUCATION

Purdue University — West Lafayette, IN
PhD in Electrical and Computer Engineering — Aug. 2021–Dec. 2026
Advisor: Prof. Qiang Qiu

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Electrical and Computer Engineering — Jan. 2019–May 2020

Nanjing University — Nanjing, China
Master of Science in Electronic Science and Engineering — Aug. 2015–Jun. 2018

Nanjing University — Nanjing, China
Bachelor of Science in Physics — Aug. 2011–Jun. 2015

RESEARCH PROJECTS

Structured Model Memory and Compositional AI — May 2025– Present
Purdue University — West Lafayette, IN

- As AI systems accumulate many specialized models and skills, studied how they can be organized as structured memory for efficient retrieval, reuse, merging, and continual updating.
- Developed structured model embedding spaces that organize task-specific models according to the knowledge and capabilities encoded in their parameters.
- Studied how relationships among tasks can be revealed directly from model parameters, enabling efficient indexing and retrieval of specialized models and skills.
- Developed methods for composing model embeddings to combine capabilities learned from different tasks.
- Explored model memory as a scalable mechanism for AI systems to retrieve, reuse, and continually update previously learned knowledge.

Related work: *Exploiting Neural Network Memory in a Structured Model Embedding Space; Approaching Effective Merging in Model Embedding Space.*

Structured Adaptation and Compositional Learning — Jan. 2023–May 2025
Purdue University — West Lafayette, IN

- Motivated by the cost and redundancy of full-model fine-tuning, studied how pre-trained parameter structure can support sparse, efficient, and compositional adaptation.
- Developed subspace-based fine-tuning methods that adapt less than 1% of model parameters while preserving reusable structure from pre-trained models.
- Extended structured adaptation across vision transformers, diffusion models, and language models, supporting downstream tasks ranging from recognition to personalized generation.
- Developed sparse representations of learned concepts and demonstrated that their components can be selectively manipulated and composed for controllable generation and in-context compositional learning.

Related work: *Large convolutional model tuning via filter subspace* (ICLR 2025)*; Coeff-Tuning: A Graph Filter Subspace View for Tuning Attention-Based Large Models* (CVPR 2025 Spotlight)*; Sparse Fine-Tuning of Transformers for Generative Tasks* (ICCV 2025)*; In-Context Compositional Learning via Sparse Coding Transformer* (NeurIPS 2025).

Structured Representations and Relations in Neural Networks — Jan. 2021–Jan. 2023
Purdue University — West Lafayette, IN

- To understand how learned knowledge is organized across neural networks, studied parameter-level representations that reveal relationships among models without relying on their original training data.

- Developed a parameter-based neural network similarity measure that enables efficient comparison of learned models and substantially reduces the computational cost of conventional data-dependent similarity measures.
- Investigated reusable filter-level structures within neural networks and developed continual learning mechanisms that transfer and recombine knowledge across tasks.
- Established the foundation for subsequent research on structured parameter spaces, model adaptation, and model composition.

Related work: *Continual Learning with Filter Atom Swapping* (ICLR 2022 Spotlight); *Inner Product-based Neural Network Similarity* (NeurIPS 2023).

RESEARCH EXPERIENCE

Surgical Instrument Identification and Defect Detection Aug. 2024– Present
Purdue University West Lafayette, IN

- Mentored junior researchers in model implementation, experimental design, debugging, validation, and performance evaluation.
- Guided the execution and iteration of experiments, helping identify technical issues and translate research ideas into reliable machine learning pipelines.
- Supported the delivery of the final systems for surgical instrument identification and automated defect detection.
- Contributions supported three patent applications currently under review.

Amazon Seattle, WA
Applied Scientist Intern May-Aug. 2024

Video Editing Using Customized Concept

- Led the project on zero-shot customized video editing in collaboration with three mentors.
- Performed a comprehensive analysis of diffusion features and identified suitable features for video editing.
- Developed an algorithm that learns personalized concepts from multiple images without additional training.
- Implemented optimal transport to maintain high feature correspondence and enhance temporal consistency.
- Published research papers at ICCV.

SELECTED PUBLICATIONS

- **Wei Chen**, Zhenyuan Dong, Qiang Qiu, *Approaching Effective Merging in Model Embedding Space*, Conference on Neural Information Processing Systems (NeurIPS), 2026.
- Vishal Purohit, **Wei Chen**, Qiang Qiu, *Blockwise Divide-and-Aggregate for Image Restoration Using Diffusion Priors*, Computer Vision and Pattern Recognition Conference (CVPR) - Findings, 2026
- **Wei Chen**, Jingxi Yu, Zichen Miao, Qiang Qiu, *In-Context Compositional Learning via Sparse Coding Transformer*, Conference on Neural Information Processing Systems (NeurIPS), 2025. (*24.52% out of 21,575 submissions*)
- **Wei Chen**, Jingxi Yu, Zichen Miao, Qiang Qiu, *Sparse Fine-Tuning of Transformers for Generative Tasks*, International Conference on Computer Vision (ICCV), 2025. (*24% out of 11,239 submissions*)
- **Wei Chen**, Huidong Liu, Yang Liu, Chien-Chih Wang, Moyan Li, Hongdong Li, Bryan Wang, *Zero-shot Customized Video Editing with Diffusion Feature Transfer*, International Conference on Computer Vision (ICCV) - Findings, 2025.
- Zichen Miao*, **Wei Chen***, and Qiang Qiu, *Coeff-Tuning: A Graph Filter Subspace View for Tuning Attention-Based Large Models*, Computer Vision and Pattern Recognition Conference (CVPR), 2025 [**Spotlight**]. (* equal contribution) (*top 2.98% out of 13,008 submissions*)
- **Wei Chen**, Zichen Miao, and Qiang Qiu, *Large convolutional model tuning via filter subspace*, International Conference on Learning Representations (ICLR), 2025. (*31.75% out of 11,672 submissions*)
- **Wei Chen***, Zichen Miao*, Qiang Qiu, *Inner Product-based Neural Network Similarity,* Conference on Neural Information Processing Systems (NeurIPS), 2023. (*26.1% out of 12,343 submissions*)
- Atul Sharma, **Wei Chen**, Joshua Zhao, Qiang Qiu, Saurabh Bagchi, Somali Chaterji, *Flair: Defense Against Model Poisoning Attack in Federated Learning*, ACM Asia Conference on Computer and Communications Security (ASIACCS), 2023.
- Atul Sharma, Joshua C Zhao, **Wei Chen**, Qiang Qiu, Saurabh Bagchi, Somali Chaterji, *How to Learn Collaboratively: Federated Learning to Peer-to-peer Learning and What's at Stake*, International Conference on Dependable Systems and Networks-Supplemental Volume (DSN-S), 2023.
- Zichen Miao, Ze Wang, **Wei Chen**, Qiang Qiu, *Continual Learning with Filter Atom Swapping,* International Conference on Learning Representations (ICLR), 2022 [**Spotlight**]. (*top 5% out of 3,391 submissions*)

- José Lezama, **Wei Chen**, Qiang Qiu, *Run-Sort-ReRun: Escaping Batch Size Limitations in Sliced Wasserstein Generative Models*, International Conference on Machine Learning (ICML), 2021. (*21.5% out of 5,513 submissions*)
- Iljoo Baek, **Wei Chen**, Zhihao Zhu, Soheil Samii, Raj Rajkumar, *FT-DeepNets: Fault-Tolerant Convolutional Neural Networks with Kernel-based Duplication*, Winter Conference on Applications of Computer Vision **(**WACV), 2021.
- Iljoo Baek, **Wei Chen**, Asish Chakrapani Gumparthi Venkat, Ragunathan Raj Rajkumar, *Practical Object Detection Using Thermal Infrared Image Sensors*, Intelligent Vehicles Symposium Workshops, 2021.
- **Wei Chen***, Kartikeya Bhardwaj*, and Radu Marculescu, *FedMAX: Mitigating Activation Divergence for Effective Federated Learning*, European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), 2020. (* equal contribution)
- Kartikeya Bhardwaj, **Wei Chen**, and Radu Marculescu, *New Directions in Distributed Deep Learning: Bringing the Network at Forefront of IoT Design*, Design Automation Conference (DAC), 2020.

MANUSCRIPTS UNDER REVIEW

- **Wei Chen**, Carter Gibbs, Colby Acton, Qiang Qiu, *Structured Model Embeddings Memory as Representations of Task Adaptations*, 2026. (under review)
- Carter Gibbs, **Wei Chen**, Colby Acton, Zhengtong Xu, Yu She, Qiang Qiu, *Action-Space Velocity Correction for Visual Consistency in Robotic Flow Policies*, 2026. (under review)
- **Wei Chen**, Colby Acton, Cheng-Yu Chu, Carter Gibbs, Qiang Qiu, *Exploiting Model Embedding Memory in Policy Learning*, 2026. (under review)
- Jingxi Yu, **Wei Chen**, Zicheng Miao, Qiang Qiu, *Markov Proposals: Iterating Visual Attention for Vision-Language Alignment*, 2026. (under review)
- Taewook Kim, **Wei Chen**, Qiang Qiu, *Addressing Textual Attention Collapse in Personalized Text-to-Image Models via Prior-Guided Attention Adjustment*, 2026. (under review)

MENTORING EXPERIENCE

Purdue University 2024– Present

Graduate and Junior Researchers

- Carter Gibbs, Colby Acton, Zhenyuan Dong, Jingxi Yu, Cheng-Yu Chu

Undergraduates

- Yuhyun Na, Aditi Anand, Michael Suo, Undergraduates from Vertically Integrated Project (VIP)

TEACHING EXPERIENCE

Lead Instructor

Job Interview in the AI-Era (2026 Summer): https://purdue-ai-bootcamp.github.io/

- Served as lead instructor for course coordination, including instructor communication, scheduling, student announcements, and overall course organization.
- Developed hands-on course materials, demonstrations, assignments, and project activities.
- Supported instruction for approximately *190* registered students across in-person and online formats.

Guest Lecturer

ECE 66200 Pattern Recognition (2026 Spring)
ECE Vertically Integrated Projects (2026)

Teaching Assistant

ECE Vertically Integrated Projects (2026 Spring)
ECE 50024 Machine Learning (2023 Spring)
ECE 57000 Artificial Intelligence (2022 Fall)

ACADEMIC SERVICE

Conference Reviewer

ECCV (2022-Present), CVPR (2022-Present), NeurIPS (2022-Present), ICML (2023-Present), ICLR (2023-Present) ICCV (2023-Present)

Program Committee

CVPR workshop: Data CV

Honors

ICCV 2025 Outstanding Reviewer (*Top 2.7% out of 11,850 reviewers*)